Varendra University

Dept. of Computer Science & Engineering (CSE)

Discussion of CO, PO, K,P,A

Course Title: **Microprocessor and Assembly Language**
Course No: **CSE 3205**

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Overview

Course Title: Microprocessor and Assembly Language
Course Code: CSE 3205
Course Type: Theory
Credits: 3
Prerequisite Knowledge: Computer Architecture
Year and Semester: 3rd Year, Summer Semester, 2026

Learning Outcomes (Theory)

COs	Description	Taxonomy domain/level	POs	WK	WP	EA	Teaching-Learning strategy	Assessment strategy
CO1	Describe Intel 8086 microprocessor Organization and Instruction sets	Cognitive/ Understand	1	4			Lecture/Slides	Mid Term
CO2	Apply flow control, shift, and rotation instructions to develop efficient program logic.	Cognitive/ Apply	2	4			Lecture/Slides	Class Test
CO3	Use memory addressing modes for efficient data access	Cognitive/ Apply	3	5			Lecture/Slides	Final

Program Outcomes (POs)

PO1. Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop solutions of complex engineering problems.

PO2. Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development (WK1 to WK4).

PO3. Design creative solutions for complex engineering problems and design systems, components or processes to meet identified needs with appropriate consideration for public health and safety, whole-life cost, net zero carbon as well as resource, cultural, societal, and environmental considerations as required (WK5).

Knowledge and Attitude Profile

Table 6.1: Knowledge and attitude Profile

	Attribute
WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences
WK2	Conceptually based mathematics, numerical analysis, data analysis, statistics and the formal aspects of computer and information science to support detailed analysis and modeling applicable to the discipline
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline
WK5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area

Knowledge and Attitude Profile

WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline
WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as professional responsibility of an engineer to public safety and sustainable development
WK8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues
WK9	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes

Complex Engineering Problems

Table 6.2: Range of Complex Engineering Problem Solving

Attribute	Complex Engineering Problems have characteristics WP1 and some or all of WP2 to WP7:
Depth of knowledge required	WP1: Cannot be resolved without in-depth engineering knowledge at the level of one or more of WK3, WK4, WK5, WK6 or WK8 which allows a fundamentals-based, first principles analytical approach
Range of conflicting requirements	WP2: Involve wide-ranging or conflicting technical, non-technical issues (such as ethical, sustainability, legal, political, economic, societal) and consideration of future requirements
Depth of analysis required	WP3: Have no obvious solution and require abstract thinking, creativity and originality in analysis to formulate suitable models
Familiarity of issues	WP4: Involve infrequently encountered issues or novel problems
Extent of applicable codes	WP5: Address problems not encompassed by standards and codes of practice for professional engineering
Extent of stakeholder involvement and conflicting requirements	WP6: Involve collaboration across engineering disciplines, other fields, and/or diverse groups of stakeholders with widely varying needs
Interdependence	WP7: Address high level problems including many components or sub-problems that may require a systems approach

Complex Engineering Activities

Table 6.3: Range of Complex Engineering Activities

Attribute	Complex activities mean (engineering) activities or projects that have some or all of the following characteristics:
Range of resources	EA1: Involve the use of diverse resources including people, data and information, natural, financial and physical resources and appropriate technologies including analytical and/or design software
Level of interactions	EA2: Require optimal resolution of interactions between wide-ranging and/or conflicting technical, non-technical, and engineering issues
Innovation	EA3: Involve creative use of engineering principles, innovative solutions for a conscious purpose, and research-based knowledge
Consequences to society and the environment	EA4: Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation
Familiarity	EA5: Can extend beyond previous experiences by applying principles-based approaches

Assessment Tools (Theory)

	Assessment Tools	Marks (%)	
Continuous Assessment (CA)	Class Participation	10%	40%
	Class Test, Assignment, Project, Discussion, Presentation	30%	
Summative Assessment (SA)	Mid-term Examination	24%	60%
	Final Examination	36%	
Total			100%

Learning Outcomes (Lab)

COs	Description	Taxonomy domain/level	POs	WK	WP	EA	Assessment strategy
CO1	Demonstrate basic input/output operations with 8086 microprocessors.	Cognitive/ Apply	1	3			Lab Test
CO2	Solve problems using decision-making and logical operation instructions	Cognitive/ Apply	2	4			Lab Test

Assessment Tools (Lab)

Assessment Tools	Marks (%)
Class Participation	10%
Continuous Assessment (CA)	40%
Lab Final	50%
Total	100%

Presentation's Evaluation (Mandatory)

Presentation's Evaluation Rubric

Criteria	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
(a) Content Quality	4	4: Clear, accurate, and relevant content	3: Mostly relevant; minor gaps	2: Adequate but lacks depth	0–1: Weak or inaccurate content
(b) Design & Organization	3	3: Well-structured, visually clean slides	2: Decent structure; minor issues	1: Cluttered or disorganized	0: Poor design/structure
(c) Delivery & Communication	3	3: Confident, engaging, clear speech	2: Fairly confident; minor hesitation	1: Reads from slides, low engagement	0: Very weak delivery
Total = 10 Marks					

Presentation’s Evaluation (Optional)

Criteria	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
(a) Content & understanding	4	4: Accurate, thorough, and demonstrates strong understanding	3: Mostly accurate; minor gaps	2: Adequate but lacks depth	0–1: Weak or inaccurate content
(b) Structure & Presentation	3	3: Well-organized, clear formatting	2: Decent structure; minor issues	1: Disorganized or unclear	0: Poor structure/formatting
(c) Timeliness & Effort	3	3: Submitted on time; clear effort and care	2: Minor delay or effort gaps	1: Late or minimal effort	0: Not submitted / very poor effort
Total = 10 Marks					